Math 4400, Fall 2016 **Quiz 3** Name: ____________________

Calculators, cell phones, or notes are not allowed.

(1) Compute $\varphi(700)$ and 3^{486} modulo 700 (i.e. 3^{486} in $\mathbb{Z}/700\mathbb{Z}$).

$700 = 2^2 \cdot 5^2 \cdot 7$ and so $\varphi(700) = \varphi(2^2) \cdot \varphi(5^2) \cdot \varphi(7) = 2 \cdot 20 \cdot 6 = 240$.

Since 3 is coprime with 700, we have $3^{240} = 1$.

Since $486 = 2 \cdot 240 + 6$, we have $3^{486} \cong_{700} 3^6 \cong_{700} 29$. (In fact $3^4 = 81$, $3^5 = 243$, $3^6 = 729 \cong_{700} 29$.)

(2) Solve $x \cong 7$ modulo 13 and $x \cong 3$ modulo 31 for some $0 \leq x \leq 402$.

We have $x = 7 + 13k$ and so $7 + 13k \cong_{31} 3$. Thus $13k \cong_{31} -4$ and hence we must find the inverse of 13 modulo 31. Running the Euclidean Algorithm we have $31 = 13 \cdot 2 + 5$, $13 = 5 \cdot 2 + 3$, $5 = 3 \cdot 2 - 1$ so $1 = 3 \cdot 2 - 5 = 13 \cdot 2 - 5 \cdot 5 = 13 \cdot 12 - 5 \cdot 31$. Thus $13^{-1} = 12$ modulo 31 and hence $k = 12 \cdot 13 \cong_{31} -4 \cdot 12 \cong_{31} -48 \cong_{31} 14$ and finally $x = 7 + 13 \cdot 14 = 189$.